Resolution Geometry: The Full Computation Loop

From Probe Histories to Stress and Curvature Tensors, and a Held-Out Test of the Gravitational Coupling

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Abstract

We execute the complete loop that connects Resolution Geometry (RG) to both tensors of gravitation:

probe histories → blind predictive state → calibrated tensor decoders → probe correction → held-out tensor forecasts → a held-out *test* of the coupling, which is never inserted.

The latent state is not a tensor. A tensor appears only when predicted, calibrated responses are decoded, and only when every admissible latent realization yields the same tensor task (task descent).

On a synthetic eccentric binary of dust blobs, two *separate* blind learners each forecast a 300-step untouched window: one sees only tidal-probe histories, the other only energy-probe histories. Their independently decoded tensors then test the coupling.

- The GR form $\operatorname{tr} E = \kappa(T_{00} + \sum_i T_{ii})$, calibrated on training data, gives $\kappa/4\pi G = 1.000000$, a measured-tensor held-out residual of 7×10^{-6}, and the verdict **GR SELECTED** at three of four observers against a trace-coupled rival (residual 6.5×10^{-3}).
- At the fourth observer the forecast budget is too wide, and the verdict is an honest **CORRESPONDENCE**.

Four boundaries are proved and witnessed:

- circular orbits make the two couplings indistinguishable;
- local stress fixes only the trace of the tidal tensor, so its traceless (Weyl-electric) part is the geometric hidden state;
- gravitationally bound sources violate $\partial T = 0$ at $O(G)$, so the linear coupling is inconsistent;
- the Newtonian gravitational stress carries exactly the missing momentum flux (relative error 2.4×10^{-5}). It is the first nonlinear term, and a pseudo-tensor, because a freely falling observer can transform g away.

Imported locks: tensor identification (461/461), the blind leap (270/270), both sides and the second-order gate (655/655, S1–S6). The truth generator embodies quasi-static linearized GR. What is tested is identifiability and discrimination, not GR in Nature.

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1 What "full computation" means here

Step	Status	Evidence
Predictive state from probe histories	executed	blind leap (24 cases); C2 (4 observers, 2 learners each)
Probe correction (acquire distinguishing probes)	executed	blind leap; C1 (tidal $1 \to 6$, energy $1 \to 10$)
Curvature decoder (tidal $\to E_{ij}$)	executed	blind leap; C1, C2
Stress decoder (energy $\to T^{\mu\nu}$)	executed	tensor identification (64 cases); C1, C2
Untouched forecasts of both tensors	executed	C2 (forecast errors 4×10^{-6} to 5×10^{-3})
Held-out coupling test (not inserted)	executed	C3: GR selected 3/4, correspondence 1/4
Coupling identifiability boundary	proved and witnessed	C4
Geometric hidden state (Weyl-electric)	proved and witnessed	C5
Conservation gate and first nonlinear term	proved and witnessed	C6
Operator selection (linear, second order)	imported	both sides (655/655), S1–S6
Full nonlinear GR, full Riemann tensor, real data	**not executed**	—

2 The loop, formally

Definition 2.1 (Predictive state and decoders)**.** From a probe history $y_{0:n}$, a learner returns a state z_n and maps

$$z_{n+1} = F(z_n), \qquad \hat{y}_n = D(z_n).$$

A *tensor decoder* acts on *calibrated predicted responses*, not on latent coordinates. The two decoders are

$$\mathcal{E}(z) = \{E : \ \|A_{\text{tidal}} \operatorname{vec} E - D(z)\| \le \epsilon\}, \qquad \mathcal{T}(z) = \{T = T^\top : \ |u_a^\mu T_{\mu\nu} u_a^\nu - D_a(z)| \le \epsilon_a\}.$$

Here A_{tidal} encodes the geodesic-deviation response $n^\top E n$ along calibrated directions, and u_a are calibrated observer four-velocities.

Theorem 2.2 (Tensor task descent)**.** *A tensor task τ has a single-valued decoder on the observable quotient if and only if $O_{\text{history}}(X_1) = O_{\text{history}}(X_2) \Rightarrow \tau(X_1) = \tau(X_2)$. The decoder is invariant under any change of latent coordinates $z \mapsto Sz$ that leaves the predictions unchanged.*

Proof. The first statement is universal factorization (RG, L2) applied to the history protocol. For the second, the decoders depend on z only through $D(z)$, and a latent similarity with $D' = DS^{-1}$ leaves $D'(Sz) = D(z)$ unchanged. □

Theorem 2.3 (Correction is monotone)**.** *For a fixed model class and fixed declared error bounds, let F_n be the set of models consistent with the first n probe histories. Then $F_{n+1} \subseteq F_n$, so the range of any task over F_n can only shrink. Strict shrinkage requires an informative probe. Enlarging the model class removes the nesting.*

3 Executed results

Generator (audit only). Two dust blobs (masses 0.05, width 1.5, $G = c = 1$) move on an eccentric orbit, $e = 0.5$, integrated with the exact blob–blob force. The stress tensor is

$$T^{\mu\nu} = \sum \rho\, u^\mu u^\nu .$$

The tidal tensor is $E_{ij} = \partial_i \partial_j \Phi$, with $\nabla^2 \Phi = 4\pi G(T_{00} + \sum T_{ii})$. This is quasi-static linearized GR; retardation and gravitomagnetism are neglected. Probe noise is 10^{-9}. Training covers samples 0–1200, validation 1200–1500, and the untouched test 1500–1800.

C1: correction loop. The probe pool is fixed before any data are seen.

- The tidal pool has 8 directions. Its rank progresses $1 \to 6$, after which the tidal tensor E is identified.
- The energy pool has 12 observers. Its rank progresses $1 \to 10$, after which the stress tensor T is identified.
- One tidal probe and one energy probe are held back and never used for fitting.

C2: blind forecasts. Per observer, two separate delay-embedded linear learners are fitted: one on the 6 tidal channels and one on the 10 energy channels. The delay is 200, and the rank is chosen by validation. Neither learner sees the generator, the orbit or the coupling.

Observer	Ranks (tidal, energy)	Tensor forecast error	Held-out probe error	Forecast budget
O_peri	(140, 140)	6.0×10^{-5}	1.2×10^{-4}	3.4×10^{-3}
O_apo	(100, 140)	5.1×10^{-3}	8.6×10^{-3}	9.7×10^{-2}
O_side	(100, 140)	1.6×10^{-3}	2.6×10^{-3}	5.3×10^{-3}
O_center	(100, 100)	3.7×10^{-6}	$< 10^{-6}$	2.7×10^{-3}

The forecast budget is hypothesis-free. It is three times the validation-window forecast error of $\operatorname{tr} E$, of T_{00} and of $\sum T_{ii}$, with models fitted on training data only.

C3: held-out coupling test. The coupling scale is calibrated on training data, using tensors decoded from the measurements. The test then uses the independently *forecast* tensors of the untouched window.

	GR: $\operatorname{tr} E = \kappa(T_{00} + \sum T_{ii})$	Rival: $\operatorname{tr} E = \kappa(T_{00} - \sum T_{ii})$
$\kappa/4\pi G$ (training)	1.000000	1.0117
Held-out residual, measured tensors	7.1×10^{-6}	6.5×10^{-3}
O_peri forecast residual	6.0×10^{-5}	6.5×10^{-3}
O_center forecast residual	2.5×10^{-6}	6.5×10^{-3}
O_side forecast residual	8.4×10^{-4}	6.3×10^{-3}
O_apo forecast residual	7.3×10^{-3}	9.6×10^{-3}

The verdict per observer is given by the budget:

- **GR SELECTED** at O_peri, O_side and O_center;
- **CORRESPONDENCE** at O_apo, where both forms lie within the forecast budget.

No observer selects the rival.

4 Boundaries, proved and witnessed

Theorem 4.1 (Coupling identifiability; C4)**.** *The GR form and the trace-coupled form are distinguishable after calibration if and only if the ratio $\sum T_{ii}/T_{00}$ varies over the data. For dust, $\sum T_{ii}/T_{00} = v^2$. On a circular orbit $|v|$ is constant, the two sources are proportional, and κ absorbs the difference: both residuals equal 2.8×10^{-10}, a CORRESPONDENCE. Discrimination needs eccentricity, pressure, or several source types.*

Proof. If $T_{00} - \sum T_{ii} = c\,(T_{00} + \sum T_{ii})$ on the data, the rival with $\kappa' = \kappa/c$ fits identically. Conversely, a varying ratio makes the two sources linearly independent on the data. □

Theorem 4.2 (Local stress fixes only the Ricci part; C5)**.** *At an observer, local stress determines $\operatorname{tr} E = 4\pi G(T_{00}+\sum T_{ii})$, which is the Ricci contraction. It does not determine the traceless part. Adding a distant heavy mass ($m = 50$ at distance 40) leaves the local T and $\operatorname{tr} E$ unchanged exactly, but changes the traceless tidal tensor by 102%. The traceless (Weyl-electric) part is the geometric hidden state, fixed by distant sources or radiation.*

Theorem 4.3 (The linear coupling needs a nonlinear term; C6)**.** *The linearized Einstein operator satisfies $\partial^\mu G^{\text{lin}}_{\mu\nu} \equiv 0$, so it requires $\partial^\mu T_{\mu\nu} = 0$. Gravitationally bound sources violate this at $O(G)$: $dP_A/dt = m_A g$.*

The Newtonian gravitational stress $t_{ij} = (g_i g_j - \frac{1}{2}\delta_{ij} g^2)/4\pi G$, integrated over a sphere around blob A, gives $-\oint t_{ij} n_j\, dS = dP_A/dt$ to relative error 2.4×10^{-5}. Conservation of $T+t$ is restored, and t is the first term of the nonlinear (Deser) completion. Because t is built from g, which a freely falling observer transforms away, it is a pseudo-tensor and not a local tensor observable. This is the operational content of the non-localizability of gravitational energy.

5 Imported locks

Battery	Result
Tensor identification (461/461)	Ten energy readings reconstruct $T^{\mu\nu}$ by polarization. Maximum component error 2.6×10^{-7}; held-out readings 4.4×10^{-7}. Planar observers give rank 6, null-only observers rank 9, and an unknown gain leaves the scale ambiguous. A quartic response fits the calibration readings but fails held-out observers. Equal-T sources differ in their hidden $l = 3$ moment.
Blind leap (270/270)	Corrected full-curvature error 1.9×10^{-6} (median), from 0.76 before correction; post-intervention assimilation 1.2×10^{-4}. An unannounced chirp is rejected. **Non-lock:** validation-selected ranks were 8–10 for a 6-dimensional generator, so minimal hidden-state recovery is not established.
Both sides (655/655)	Conservation plus descent select the linearized Einstein operator at second order. Hidden values are separated from null values; source and gravitational observers are tested.

Battery	Result
Second-order gate (S1–S6)	The fourth-order compatible space is 2-dimensional. The Stelle potential, the ghost/scalar split and hidden-state counts 2/3/7/8 follow; H-min selects second order; data lock it above a declared scale.

6 Final gate ledger

Gate	Status	Next
Tensors from probe histories	**computed** (synthetic)	real calibrated probes
Coupling (linear, second order)	selected; **tested on held-out data**	physical data with varying stress/energy ratio
Coupling scale G	calibrated (unit anchor)	—
Weyl-electric hidden state	typed; requires nonlocal data	tidal histories, waves
Weyl-magnetic part and spatial Riemann	not reconstructed	gyroscope and moving-probe panels
Nonlinear completion	first term computed (pseudo-tensor)	order-by-order bootstrap
Source microdynamics	declared (dust, free fall)	—
Minimal hidden-state recovery	**not established** (blind leap)	minimal-realization learners
Real data (H1/L1)	blocked (no detector operator)	detector response model
Full GR from RG	**not established**	—

7 Battery

`rg_full_loop.py` (NumPy, SciPy; seed 20261101) runs C1–C7 in about 9 s. Two canaries are rejected by computed criteria:

- nine energy observers identify T;
- the rival passes the eccentric test.

The imported batteries are included unchanged, with their reports and verifiers.

Limits.

- The generator embodies quasi-static linearized GR, so the coupling test validates identifiability and discrimination, not GR in Nature.
- The blobs are rigid and the motion is slow ($v \lesssim 0.14$).
- Forecast budgets are heuristic (three times the validation error).
- The battery is not blind.
- No real data are used.

8 Non-claims

1. RG does not derive GR here.
2. The latent coordinates are not tensor components; tensors appear only through calibrated decoders.
3. Curvature recovery does not recover stress without a coupling, and stress does not recover curvature (C5).
4. The trace-coupled rival is a declared comparison form, not a complete alternative theory.

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